Shiham Chowdhury

Shiham | Shiham Chowdhury | shiham@umich.edu | Portfolio

Education

University of Michigan — **May 2027 (Expected)**
Bachelor of Science in Electrical Engineering — *Ann Arbor, MI*

Technical Skills

Languages: Java, Javascript, C++, Python, MATLAB
Developer Tools: Visual Studio, Jupyter Notebook, LTSpice, React, Git
Hardware: Soldering, Oscilloscope, PCB Design, Network Analyzer, Arduino, Raspberry Pi

Experience

eTitle | *Troy, MI* — May 2025 - August 2025

- Helped develop a web-based foreclosure management system designed to ease user convenience
- Tested updated merge templates using sample foreclosure cases to ensure proper functionality
- Optimized merge template scripts by removing hard-coded elements to replace outdated data

Wolverine Solar Car Team | *Ann Arbor, MI* — August 2023 - present

- Prevented battery cells from exceeding 40 °C under high load by implementing a PID-controlled PWM fan system, doubling projected lifespan and reducing risk of thermal runaway.
- Improved cooling system by designing a low-pass filter in MATLAB, reducing false fan activations by 30% and cutting temperature fluctuations by 20% for stable PID control.
- Tested battery cooling system by running trials through varied terrain and weather, reducing thermal derating events by 25% and maximizing peak power uptime during endurance runs.

Projects

Environment Monitoring Device | *Arduino IDE, Amazon Web Services (AWS)* — Link to Project

- Built an IoT system with ESP32 and DHT22 sensors to collect real-time climate data, enabling accurate monitoring of environmental conditions.
- Ensured reliable data access by integrating AWS IoT Core for secure transmission and AWS S3 for scalable storage of continuous climate readings.
- Developed a React front-end application that displays data from S3 using EC2 MATLAB scripts, improving user accessibility and real-time insight.

Word Occurrence Prediction Algorithm | *C++* — Link to Project

- Developed a C++ algorithm to predict word occurrence probabilities from user input, enabling analysis of language patterns.
- Improved prediction accuracy by applying machine learning algorithms to categorize word pairs and refine probability models.
- Employed recursive binary search trees to efficiently parse and manage learned data, facilitating quick computations of log-probability scores.

Euchre Game | *C++* — Link to Project

- Designed and implemented a fully functional Euchre game in C++ that supports up to 4 players.
- Integrated comprehensive game logic to enforce Euchre rules and maintain accurate game states.
- Implemented strategic decision-making algorithms based on game state to simulate CPU opponents.

Temperature-Controlled Fan | *Arduino IDE, Arduino Uno* — Link to Project

- Built an Arduino-controlled fan that monitors temperature and humidity using a DHT11 sensor to display on a 16x2 LCD screen for real-time feedback.
- Implemented logic to activate DC motor-powered fan based on temperature readings or when ultrasonic sensor detects an object getting too close, minimizing excess fan operations.